

1 Balance over all energies

$$\begin{aligned}\Sigma_t(E)\phi(E) &= \int_0^\infty \Sigma_s(E' \rightarrow E)\phi(E') \, dE' \\ &\quad + \chi(E)s_f''' + s_{\text{ext}}'''(E), \\ s_f''' &= \int_0^\infty \nu(E)\Sigma_f(E)\phi(E) \, dE.\end{aligned}$$
$$R_x = \int_0^\infty \Sigma_x(E) \phi(E) \mathrm{d}E, \quad s = \int_0^\infty s_{\text{ext}}'''(E) \mathrm{d}E.$$
$$R_t = R_s + s_f''' + s \quad \Rightarrow \quad \boxed{R_a = \underline{\hspace{2cm}}}.$$

Checkpoint

2 Preserve the reaction rate

Flux-weighted average (FNRP 3.38)

$$\bar{\Sigma}_x = \frac{\int_0^\infty \Sigma_x(E) \phi(E) dE}{\phi_{\text{tot}}}, \quad R_x = \bar{\Sigma}_x \phi_{\text{tot}}.$$

$$\overline{\nu \Sigma_f} = \frac{\int_0^\infty \nu(E) \Sigma_f(E) \phi(E) dE}{\phi_{\text{tot}}}.$$

Nonmultiplying, source-driven limit

$$\bar{\Sigma}_a \phi_{\text{tot}} = s \quad \Rightarrow \quad \phi_{\text{tot}} = \underline{\hspace{2cm}}.$$

Check: $\bar{\Sigma}_g$ has units cm^{-1} and ϕ_{tot} has units $\text{cm}^{-2} \text{s}^{-1}$.

g	Interval $[E_g^L, E_g^U]$	Shape $h_g(E)$
T	$[10^{-3}, 1]$	$Ee^{-E/\theta}$
I	$[1, 10^5]$	$1/[\Sigma_i(E)E]$
F	$[10^5, 10^7]$	$\chi(E)/\Sigma_i(E)$

$$C_g^{-1} = \int_{E_g^L}^{E_g^U} h_g(E) \mathrm{d}E, \quad w_g(E) = C_g h_g(E),$$

$$\int_{E_g^L}^{E_g^U} w_g(E) \mathrm{d}E = 1, \quad \phi_g = \int_{E_g^L}^{E_g^U} \phi(E) \mathrm{d}E.$$

$$\phi(E) \approx \phi_g w_g(E)$$

Shape versus amount

4 Combine the range averages

$$\bar{\Sigma}_{x,j} \approx \int_{E_g^L}^{E_g^U} \Sigma_x(E) w_g(E) dE.$$

Add rates before dividing by total flux

$$R_x \approx \bar{\Sigma}_{x,T} \phi_T + \bar{\Sigma}_{x,I} \phi_I + \bar{\Sigma}_{x,F} \phi_F,$$

$$\phi_{\text{tot}} \approx \phi_T + \phi_I + \phi_F.$$

$$\bar{\Sigma}_x \approx \frac{\bar{\Sigma}_{x,T}\phi_T + \bar{\Sigma}_{x,I}\phi_I + \bar{\Sigma}_{x,F}\phi_F}{\phi_T + \phi_I + \phi_F}$$

$$\bar{\Sigma}_v \approx \underline{\hspace{10cm}}$$

5 Three useful spectrum averages

Fission-spectrum average. If Σ_f is approximately constant over the fast range,

$$\bar{\sigma}_{x,F} \approx \frac{\int_{\mathcal{E}_F} \sigma_x(E) \chi(E) dE}{\int_{\mathcal{E}_F} \chi(E) dE}.$$

The denominator is one only when the chosen interval contains all of the normalized fission spectrum.

Maxwellian average. Extend the ideal Maxwellian to all $E > 0$:

$$w_T(E) = \frac{E e^{-E/\theta}}{\theta^2}, \quad \int_0^{\infty} E e^{-E/\theta} dE = \theta^2.$$

For a $1/v$ cross section, $\sigma_x(E) = \sigma_0 \sqrt{E_{\text{ref}}/E}$,

Average a $1/v$ absorber

$$\bar{\sigma}_{x,T} = \frac{\sigma_0 \sqrt{E_{\text{ref}}}}{\theta^2} \int_0^{\infty} E^{1/2} e^{-E/\theta} dE$$

$$= \frac{\sigma_0 \sqrt{E_{\text{ref}}}}{\theta^2} \int_0^{\infty} E^{1/2} e^{-E/\theta} dE = (\sqrt{\pi}/2) \theta^{3/2}.$$

Use $\int_0^{\infty} E^{1/2} e^{-E/\theta} dE = (\sqrt{\pi}/2) \theta^{3/2}$.

Resonance integral. For $1/E$ weighting on $[E_t, E_u]$,

$$I_x(E_t, E_u) = \int_{E_t}^{E_u} \frac{\sigma_x(E)}{E} dE, \quad \bar{\sigma}_{x,I} = \frac{I_x}{\ln(E_u/E_t)}.$$

I_x has cross-section units. Its bounds and any subtraction of a $1/v$ contribution must match the data definition.

Do not mix weighting assumptions

The NR average uses the weight $1/[\Sigma_f(E)E]$. An ordinary $1/E$ resonance integral does *not* include that flux depression.

6 Source-driven water tank

Treat Prof. McNeil's water tank as infinite and homogeneous. Trace ^{252}Cf supplies $s = 10 \text{ cm}^{-3} \text{ s}^{-1}$; ignore neutron interactions with the trace source. Assume $\phi_T = \phi_I = \phi_F$. For water molecule density N_w ,

$$\bar{\Sigma}_a = \frac{N_w}{3} \sum_{j=T,I,F} (2\bar{\sigma}_{a,j}^H + \bar{\sigma}_{a,j}^O).$$

Use microscopic cross sections in cm^2 ($1 \text{ b} = 10^{-24} \text{ cm}^2$). Find

$$\phi_{\text{tot}} = \text{_____}, \quad \phi_j = \text{_____}.$$

7 Source-free multiplication

With no external source, physical steady-state balance is

$$\bar{\Sigma}_a \phi_{\text{tot}} = \bar{\nu} \Sigma_f \phi_{\text{tot}}.$$

For a *nonzero* flux, production must equal absorption. This is the critical condition, but it does not determine the flux magnitude.

Introduce the infinite-medium eigenvalue

Scale only fission production to obtain a stationary eigenproblem:

$$\bar{\Sigma}_a \phi_{\text{tot}} = \frac{1}{k_{\infty}} \bar{\nu} \Sigma_f \phi_{\text{tot}}, \quad k_{\infty} = \frac{\bar{\nu} \Sigma_f}{\bar{\Sigma}_a}.$$

With the consistent eigen-spectrum this is the eigenvalue; a prescribed approximate spectrum gives an estimate.

Interpret the ratio

Classify the infinite system:

$$k_{\infty} < 1 : \text{_____},$$

$$k_{\infty} = 1 : \text{_____},$$

$$k_{\infty} > 1 : \text{_____}.$$

Does doubling $\phi(E)$ change k_{∞} ? Does it change R_f ?

8 U-235 dissolved in water

Replace the external source by dissolved, pure ^{235}U . Ignore other solute constituents and leakage. Define

$$r = \frac{N_U}{N_w}, \quad \phi_T = \phi_I = \phi_F.$$

For each nuclide, average over the *same mixture spectrum*; with equal range fluxes, $\bar{\sigma}_x^i = \frac{1}{3} \sum_j \bar{\sigma}_{x,j}^i$. Only uranium contributes fission production in this model:

$$\bar{\Sigma}_a = N_w (2\bar{\sigma}_a^H + \bar{\sigma}_a^O + r\bar{\sigma}_a^U),$$

$$\bar{\nu} \Sigma_f = N_w r \bar{\nu} \sigma_f^U.$$

Form production divided by absorption

$$k_{\infty} \approx \frac{r \bar{\nu} \sigma_f^U}{2\bar{\sigma}_a^H + \bar{\sigma}_a^O + r\bar{\sigma}_a^U}.$$

Why does N_w cancel? What is being held fixed when this expression is used to vary r ?

Takeaway

Changing composition generally changes the spectrum and its averages. Equal thermal, intermediate, and fast fluxes are an *assumption for this example*, not a prediction.